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PALAEOMAGNETIC STUDIES OF LOWER JURASSIC ROCKS IN ENGLAND AND WALES

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Paleomagnetic data from late Lower Jurassic sediments from Dorset, Gloucestershire and Yorkshire confirm the existence of at least one interval of reversed geomagnetic polarity in Upper Toarcian time (about 175 m.y. ago). The mean pole position calculated from these data lies at 50°N, 37°E, but the field represented by this pole is believed to have been non-axial because of its incompatibility with Mesozoic faunal, palaeoclimatic and palaeomagnetic evidence throughout western Europe.

Unstable remanent magnetization found in a large number of samples of various lithologies is shown to have originated during drilling. It is tentatively identified as partial TRM acquired during dissipation of heat from the drill tip. Alternative causes such as superficial contamination and vibration do not explain the majority of the observations.

1. Introduction

The Lower Jurassic period is of particular interest because the present expanding ocean basins appear to have originated about that time and the oldest surviving large fragments of ocean floor are of about that age. This period therefore offers the potential of using palaeomagnetism to chart the early stages of continental break-up and ocean formation, and is the earliest geological time at which geomagnetic polarity information derived on land and from the ocean-floor can be cross-checked.

Unfortunately the land-based palaeomagnetic record at this time is rather sparse, the age of most of the rocks which have been studied are controlled by radiometric rather than stratigraphic dating, and correlation of the two time scales is not particularly refined at that period [1]. The extensive Jurassic sedimentary sequence in western Europe and particularly in Britain might be thought ideal for study, being exceptionally well dated stratigraphically by ammonite zonation, but the lithologies which are present – mainly non-ferruginous limestones, sandstones and clays – are not amongst those commonly regarded as favourable for palaeomagnetic study. Hence the present investigation was prompted by a need to bridge an unusually large gap in the palaeomagnetic record for western Europe by delineating a portion of the apparent polar wander path which has hitherto been inferred by interpolation between Upper Triassic and Lower Cretaceous data. Moreover it was encouraged because it has become clear in recent years that weakly magnetised sediments can in fact carry stable and original remanence which is detectable by sensitive magnetometers, and by the fact that two early studies in this area [2,3] had yielded coherent, if limited, results. An extra incentive was that one of these early results had been seized on, possibly prematurely, to define an alleged geomagnetic polarity event in the early Jurassic [4] which merited more detailed study.

This study has been concentrated in three areas which will be described in turn. From the Cotswolds to Dorset the sequence is mainly marine, and four groups of ferruginous sands of Upper Pliensbachian to basal Bajocian age were sampled in quarries and road cuts. In most cases cores were drilled on outcrop at horizons extending no more than a few metres laterally

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and 10 cm vertically. Each core is treated as a specimen in our analysis and each horizon is regarded as providing a representative palaeomagnetic observation. Likewise in the few cases where block sampling was substituted for drilling, each block is regarded as providing a representative observation which is estimated from laboratory measurements on a number of core specimens cut from it on a drill press. Collections are confined to a few localities, but within each outcrop each horizon, as defined above, is representative of a distinct stratigraphical level and hence is taken as the basic unit for statistical analysis.

On the North Yorkshire coast the same time span is represented by a mainly argillaceous sequence containing calcareous concretions and thin calcareous interbeds. These were sampled at 27 sites spread along 3 km of coastline by core sampling at outcrop. Inland, at Rosedale, an ironstone of uppermost Toarcian age was sampled and ten specimens were cut from four blocks for measurement.

Jurassic rocks do not outcrop in Wales, but a scientific exploration borehole sunk by the Institute of Geological Sciences at Llanbedr (Mochras Farm) in 1969 [5] penetrated 1305 m of Lower Jurassic beneath 601 m of Tertiary to Recent cover. The whole borehole was cored, and from these large diameter cores 414 small 2.5 × 2.5 cm cylindrical specimens were cut for palaeomagnetic study. These were taken mainly in pairs for consistency checking, at 224 levels unevenly distributed throughout the length of the borehole. Knowing the borehole to be close to vertical, but without azimuthal orientation of the core, consistent NRM would be observed by consistency of measured inclination but randomly distributed declination relative to our cut specimens. This unusually thick Lower Jurassic sequence held the prospect of extremely detailed palaeomagnetic information but these hopes were not fulfilled. Instead a "spurious" magnetization acquired during drilling was identified throughout, which completely masked any pre-existing remanence. Hence particular attention is paid throughout this paper to the question of drilling-imposed remanences.

Girdler's [2] study in the Cotswolds, Somerset and Dorset covered four formations, three of which have been resurveyed in the present study, and the earlier results substantially confirmed. The remaining one, the Midford Sands, has not been repeated but is discussed in the concluding section. Nairn [3] briefly reported results from limestones in Scotland. One of us (J.T.S.) made a more extensive collection in both Sutherland and Skye with Dr. J.D.A. Piper, but these have not yet been investigated in the laboratory.

2. Palaeomagnetic measurements

2.1. Dorset and the Cotswolds

Pennard Sands (Upper Pliensbachian, Middle Lias) outcrop in a roadcut at Sandford Orcas where six horizons (29 cores) were sampled in about 5 m of yellow micaceous sands, which are mainly fine grained and well bedded, with calcareous concretions. A further group of twelve specimens was taken from an immediately overlying 40–50 cm hard calcareous sandstone band. NRM directions were tightly grouped around a mean only slightly steeper than the present geomagnetic field (Table 1). Intensities were typically of the order 10^{-6} gauss. During a.f. demagnetization direction changed little in peak fields up to 500 Oe but treatment in higher fields resulted in large irregular changes. Intensity decreased approximately linearly up to 500 Oe and then oscillated irregularly. Fisher analysis of seven specimens magnetically cleaned at 300 Oe gave mean direction and precision indistinguishable from the total NRM of the whole collection (Table 1). Thermal demagnetization was performed on only a few specimens and typically revealed poor stability.

The Cotswold Sands (Upper Toarcian, Upper Lias) consist of yellow micaceous fine-grained sands with calcareous concretions. They were sampled at the two localities studied by Girdler [2] (D.V. Ager, personal communication). At Crickley Hill four horizons within a stratigraphic thickness of 1.5 m were sampled together with two horizons in overlying Scissum Beds (basal Bajocian). At Wootton-under-Edge the uppermost Toarcian Cephalopod Bed intervenes between the Cotswold Sands and Scissum Beds. It may be, then, that the Wootton section of Cotswold Sands is slightly the older; 13 horizons were sampled in 6 m of its exposure, and three more in the Cephalopod Bed. Reversed NRM was observed throughout the Wootton section, in accord with Girdler's result; detailed confirmation of its stability is therefore important (Fig. 1a–d). Directional stability and linear reduction of

TABLE 1

Palaeomagnetic data of the present study, in approximate stratigraphic order

Formation	Locality (N.G.R.)	Lat./long.	Total NRM						Partial demagnetization						Polarity	Palaeomagnetic pole[1] (lat./long.)
			horizons (*N*)	cores	R_N	k	α_{95}	D, I	treatment	specimens (*N*)	R_N	k	α_{95}	D, I		
Bridport Sands	Burton Bradstock (SY487891)	50°42′N, 2°44′W	8	73	7.86	51	7.8	25, +66	a.f. 300 Oe	6	5.72	18	16	82, +78	N	52N, 35E (*b*)
Yeovil Sands	Babylon Hill (ST580160)	50°57′N, 2°36′W	7	79	6.83	36	10.2	14, +65	a.f. 300 Oe	7	6.48	11	19	25, +59	N	–
Cephalopod Bed	Wootton-under Edge (ST735994)	51°39′N, 2°21′W	3	10	2.45	3.6	–	308, −54	a.f. 300 Oe[4]	3[3,6]	2.72	7	–	285, −83	R	46N, 17E (*c*)
Cotswold Sands	Crickley Hill (SO930160)	51°51′N, 2°6′W	6[2]	19	5.54	11	21.2	347, +51	a.f. 300 Oe	3	2.93	28	23.7	39, +44	N	–
	Wootton-under-Edge (ST735994)	51°39′N, 2°21′W	13	43	12.56	27	8.1	290, −70	a.f. 300 Oe[4]	13[3,5]	12.74	46	6.1	260, −74	R	47N, 43E (*a*)
Rosedale Iron Ore	Rosedale (SE706989)	54°23′N, 0°54′W	4[3]	10	3.99	500	4.0	222, −76	thermally stable to 600°C						R	67N, 46E (*d*)
Pennard Sands	Sandford Orcas (ST621208)	50°59′N, 2°32′W	7	41	6.85	41	9.5	347, +69	a.f. 300 Oe	7	6.83	35	10.4	4, +65	N	–

[1] Key to poles illustrated in Fig. 2.
[2] Includes two horizons in Scissum Beds.
[3] Number of samples given unit weight in the analysis.
[4] Cores cut with steel drill magnetically cleaned, total NRM of cores cut with phosphor-bronze drill also included.
[5] 37 specimens involved.
[6] 5 specimens involved.

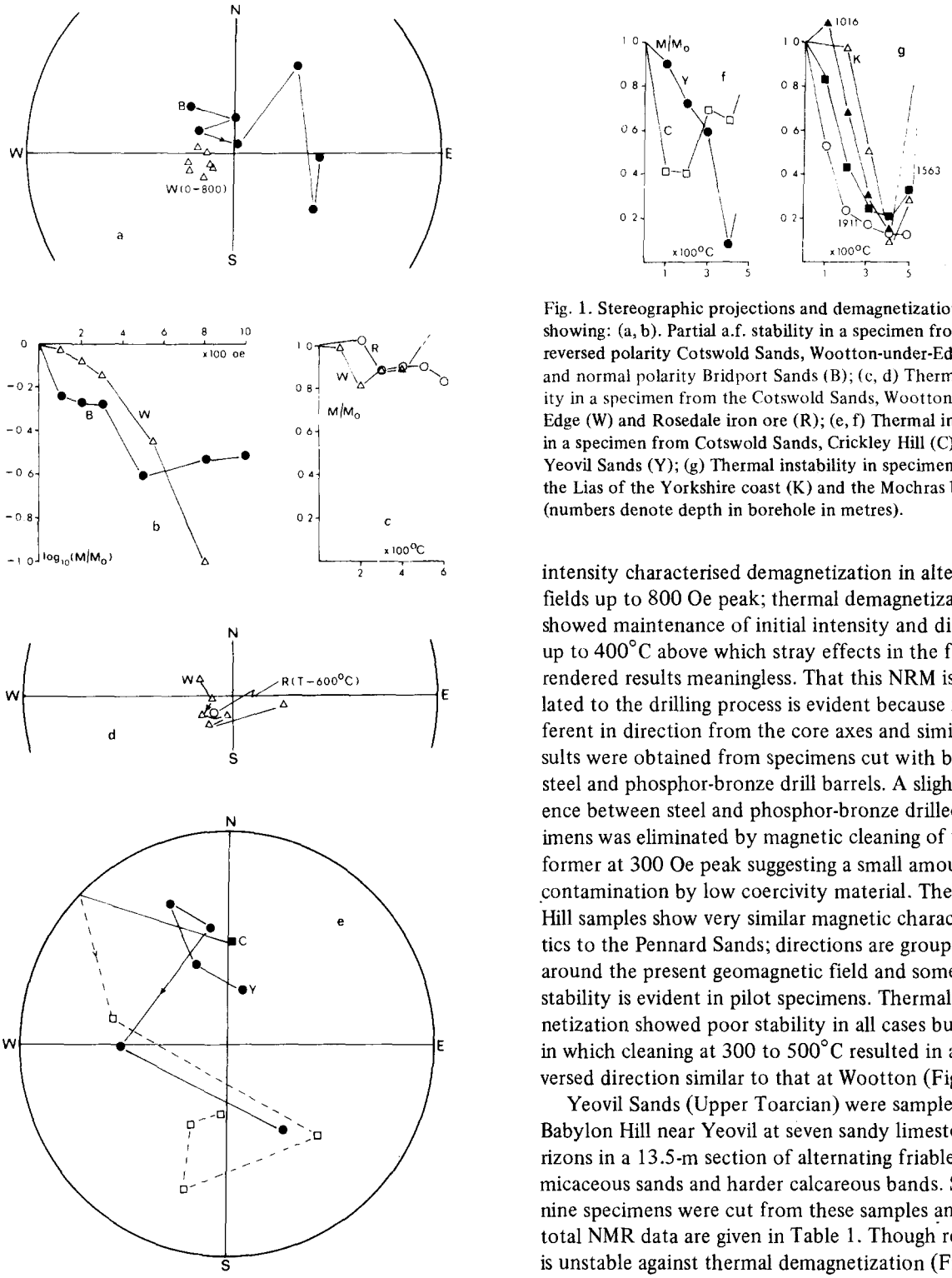


Fig. 1. Stereographic projections and demagnetization curves showing: (a, b). Partial a.f. stability in a specimen from the reversed polarity Cotswold Sands, Wootton-under-Edge (W) and normal polarity Bridport Sands (B); (c, d) Thermal stability in a specimen from the Cotswold Sands, Wootton-under-Edge (W) and Rosedale iron ore (R); (e, f) Thermal instability in a specimen from Cotswold Sands, Crickley Hill (C) and Yeovil Sands (Y); (g) Thermal instability in specimens from the Lias of the Yorkshire coast (K) and the Mochras borehole (numbers denote depth in borehole in metres).

intensity characterised demagnetization in alternating fields up to 800 Oe peak; thermal demagnetization showed maintenance of initial intensity and direction up to 400°C above which stray effects in the furnace rendered results meaningless. That this NRM is unrelated to the drilling process is evident because it is different in direction from the core axes and similar results were obtained from specimens cut with both steel and phosphor-bronze drill barrels. A slight difference between steel and phosphor-bronze drilled specimens was eliminated by magnetic cleaning of the former at 300 Oe peak suggesting a small amount of contamination by low coercivity material. The Crickley Hill samples show very similar magnetic characteristics to the Pennard Sands; directions are grouped around the present geomagnetic field and some a.f. stability is evident in pilot specimens. Thermal demagnetization showed poor stability in all cases but one, in which cleaning at 300 to 500°C resulted in a reversed direction similar to that at Wootton (Fig. 1e, f).

Yeovil Sands (Upper Toarcian) were sampled at Babylon Hill near Yeovil at seven sandy limestone horizons in a 13.5-m section of alternating friable yellow micaceous sands and harder calcareous bands. Seventy-nine specimens were cut from these samples and the total NMR data are given in Table 1. Though remanence is unstable against thermal demagnetization (Fig. 1e, f)

TABLE 2

Magnetizations believed to have originated during drilling

Formation	Locality (N.G.R.)		Lat./long.	Horizons	cores (N)	R_N	k	α_{95}	D, I[1]	Comments
Upper Lias	N. Yorkshire (NZ783188–800178)		54°42′N,0°48′W	27	128	96.5	4	8.0	356, −8	a.f. and thermal demagnetization of selected specimens show instability
Tertiary		Llanbedr		33	62	20.4	1.4	26.2	176, +15	borecore drilled in direction (180,0)
Middle Lias		(Mochras Farm)		54	105	81.0	4.3	7.5	356, +30	borecore drilled in direction (−, +90)
Lower Lias	(to 1128 m)	Borehole	52°49′N,4°10′W	55	103	77.5	4.0	8.0	350, +41	unstable on partial thermal demagnetization
Lower Lias	(from 1128 m)	(SH553259)		77	139	117.2	6.3	5.1	348, +45	
Trias				5	5	4.85	27	14.8	348, +44	
Eocambrian sediments, Finmark (Norway)			70° N, 28° E	9[2]	89	8.61[2]	21	11.6	356, +11	stable in alternating fields of 800 Oe (peak)

[1] Directions specified relative to core axes, with direction of drilling defined as $D = 0°$, $I = 0°$.
[2] Unit weight given to each horizon.

it is little changed by a.f. cleaning at 300 Oe peak field after which seven selected specimens, one from each site, yielded directions indistinguishable from the total NRM and from the present geomagnetic field.

Eight horizons in hard calcareous sandstone bands interbedded with more friable and finer sands in an 8-m section of Bridport Sands (Upper Toarcian) at Burton Bradstock were sampled. Total NRM results from 73 specimens are summarised in Table 1. The bulk of the NRM has low coercivity (Fig. 1a, b) and critical blocking temperature, but directions after a.f. cleaning at six of the eight sites at 300 Oe were grouped around the direction $D = 82°$, $I = +78°$.

2.2. *North Yorkshire*

NRMs of the bulk of this collection were very weak, typically less than 10^{-7} gauss rendering partial demagnetization impracticable in most cases. A single example is given in Fig. 1g. The key to their interpretation is that NRM is preferentially aligned along the axis of the cylindrical cores (Table 2). Hence this remanence, whose properties and origin are discussed in section 3.1 is probably related to the drilling process. The observations that most site mean directions were statistically significant but that they had no common feature except positive inclination, are to be explained by non-randomness of direction of drilling at individual outcrop sites.

By contrast the NRM in ten specimens (four hand samples) of ironstone of uppermost Toarcian age at Rosedale was stronger and unrelated to the direction of the core axes. Directions were initially well grouped with reverse polarity (Table 1) and changed little during thermal demagnetization up to 600°C (Fig. 1c, d). In two of the three test specimens intensity changed little during this treatment, while in the third it was reduced to half. Trace hematite is suspected as the remanence carrier in these chamosite siderite oolites.

2.3. *Llanbedr (Mochras Farm) borehole*

NRM in all the borehole samples was less than 10^{-6} gauss. That it was predominantly acquired during the drilling of both the original borehole and the subsequent palaeomagnetic sampling is evident from the following

observations and the summary information in Table 2. The Tertiary section was sampled by coring antiparallel to the original borehole direction. NRM is scattered but non-random, the mean being *along* the borehole axis (which with the convention used would be the direction $D = 180°, I = 0°$. The boring rather than subsequent specimen cutting is therefore the dominant cause of remanence. The Jurassic and Triassic sections were sampled by cutting at right angles to the borehole axis in unknown and presumably random original azimuth. NRM lies intermediate between the borehole axis ($I = +90°$) and the axis of our specimens ($D = I = 0°$) showing that both processes contribute to NRM. Thermal demagnetization showed poor stability (Fig. 1g).

3. Interpretation

3.1. Magnetization related to drilling

Explanation of such remanences must take account of the following observations.

(1) Remanence is aligned with the direction of drilling rather than the local geomagnetic field in which they were drilled.

(2) The magnetic field of the drill barrel is only of the order of 10 mOe near the drill tip.

(3) In the Mochras and Yorkshire rocks the effect is similar in cores cut with steel and phosphor-bronze drill barrels and cannot be removed by acid leaching or scrubbing of the core surfaces. Only in the Wootton rocks was such a surface contamination effect found, and it was removable by a.f. cleaning at 300 Oe (peak).

(4) With that one exception, the effect has only been identified in rather hard rocks in which drilling is either slow or generates heat due to persistent failure of the lubricant.

(5) The effect has been found in impermeable phyllites and quarzites of Eocambrian age from Finnmark (J.C.B., unpublished studies of collections by Dr. H.G. Reading and colleagues summarised in Table 2).

(6) In the lithologically uniform Jurassic sequence of the Mochras borehole the effect from drilling the original hole increases with depth, because grouping of NMR increases and the mean moves progressively towards the down-hole direction. Properties of specimens cut from the heart and the periphery of the borecores are indistinguishable.

(7) The NRM in the Mochras and Yorkshire rocks had low critical blocking temperature. In the harder Finnmark rocks intensity was greater and was not completely removed by a.f. demagnetization at 800 Oe (peak).

(8) Reflected light microscopy undertaken by Dr. J.M. Ade-Hall revealed no opaques whatsoever in samples from Finnmark. One section from Mochras contained a single euhedral magnetite grain about 100 μ diameter. The other rocks contain sparse unidentified ultrafine opaques.

An origin as partial TRM resulting from heat dissipation at the drill tip satisfies all the observations except the apparent alignment with the drill axis (the field due to the drill being two orders of magnitude less than the local geomagnetic field when measured at room temperatures in the laboratory). If the drill barrel has greater effect when it is itself heated during coring then this may be the answer; otherwise it seems that a different common cause exists which has yet to be identified.

Alignment of the remanence with drilling direction despite the apparent weak magnetic field due to the drill, seems to point to mechanical axial orientation of the remanence carrier. Yet surface contamination and grain reorientation due to vibration of the drill which have been proposed as mechanisms by Burmester [6] and Kuster [7] respectively do not satisfy the majority of the evidence; nor does reorientation by percolating drilling fluid.

3.2. Magnetizations close to the present geomagnetic field

NRM of normal polarity, in direction indistinguishable from the present geomagnetic field, has been found in the Yeovil Sands, and is the dominant component in the Bridport Sands, as reported by Girdler [2]. The same is found in the Pennard Sands. These rocks have typically poor stability against thermal and a.f. demagnetization and cannot be judged with any confidence as carriers of Jurassic remanence. However, the possibility that the geomagnetic field may have been in this direction in part of Jurassic time is entertained in the following section so it remains conceivable that it is being observed in these rocks.

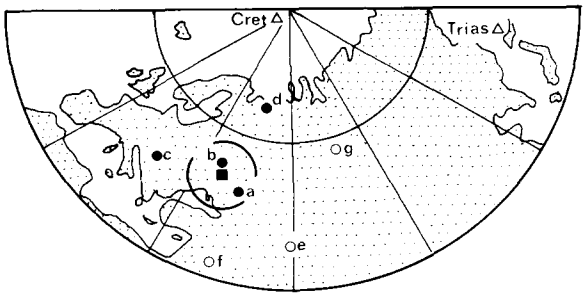


Fig. 2. Polar stereographic outline sketch map of Eurasia, showing palaeomagnetic poles *a–d* of Table 1, together with their mean (see text) as a solid square with its circle of 95% confidence. *e*, *f* and *g* are the previously published poles from Cotswold Sands at Crickley Hill [2], Midford Sands [2] and Scottish Middle Jurassic sediments ([3], as recalculated in [12]), respectively. Mean Triassic and Cretaceous poles for northern Europe [11] are added for comparison. Solid symbols: this study. Open symbols: published data.

3.3. Magnetizations of probable Jurassic age

The reverse NRM in Cotswold Sands and the Rosedale iron ore are clearly not of geologically recent origin. In view of their greater stability in comparison with all other rocks in the present study and the virtual identity of their stratigraphic ages, it is probable that they both record the same reversed polarity interval in late Toarcian time. The beginning of this polarity interval cannot be determined with certainty because the only older rocks studied were the Pennard Sands of normal polarity but doubtful stability. The end of the interval is probably defined by the normal polarity of NMR in the Midford Sands [2] and a.f. cleaned specimens of Bridport Sands. Both these formations are of late Toarcian age, younger than the Cotswold Sands; they both yield pole positions significantly different from the present geographic and geomagnetic poles but similar to those of the Wootton and Rosedale reversed rocks (Fig. 2). There is a possiblity that the Cephalopod Bed at Wootton is recording a younger reversed polarity interval close to the Lower/Middle Jurassic boundary. This thin bed is a highly condensed sequence extending through the time span of the Midford and Bridport Sands and into the lowest Bajocian. It is more likely that its matrix was lithified at that later time than that it is sedimentologically and palaeomagnetically contiguous with the underlying Cotswold Sands. The existence of the reversed polarity which gave its name to the Cotswold Reversed Zone of McElhinny and Burek [4] is thus confirmed. The R-N transition between this and the Midford Sands is consistent with its correlation with the R-N transition in the Stormberg lavas in south Africa by those authors. Even if the Cephalopod Bed does record another reversed event it is doubtful if it is to be correlated with McElhinny and Burek's next such event (Mateke Zone). Correlation with Pechersky's polarity scale for the Mesozoic of the U.S.S.R. as reported by Creer [8] is equally hazardous; the compatibility of the puclished scales is in doubt [8] and already various dubious correlations between them have been claimed [8, 9]. It is conceivable that one or both of the brief Toarcian reversed events, illustrated but unnamed in the paper by Creer [8], may coincide with the British observations. Time scales based on marine magnetic anomalies, which might clarify the situation, currently stop short of the Lower Jurassic [10].

The best estimate of the Toarcian palaeomagnetic pole relative to Britain from the present study is obtained by combining the data from the reversed Wootton and Rosedale rocks with the normal a.f. cleaned Bridport data. Normalising polarities and giving unit weight to each sample ($N = 26$) yields: $D = 79°$; $I = +77°$ with $k = 27$, $\alpha_{95} = 5.5°$ corresponding to a palaeomagnetic pole at 50°N, 37°E, as illustrated by a square in Fig. 2. This is in fair agreement with previously published data (Fig. 2) but is to be preferred because of the laboratory evidence of stability. This pole does not lie in direct line between the mean Triassic and Cretaceous poles for western Europe computed by McElhinny [11]; four explanations come to mind. At times when the palaeomagnetic axis is close to the present rotation axis there is always the risk of bias against any data which are statistically indinstinguishalbe from the present geomagnetic field unless reversals or strong stability are evident; hence it is conceivable that we have wrongly discounted the results from the Pennard and/or Yeovil Sands. On the other hand their instability demands their exclusion [2, 13]. Failure to average palaeosecular variation of periods up to 10^5 years could bias the result, but in this study at least two ammonite zones have been covered. The Mesozoic polar wander path for Europe could genuinely have a zigzag shape. The contemporary path for North America shows some rapid shifts, but the path is not the same shape and adjustment of

Europe and North America into plausible Jurassic relative positions accentuates the difference between them. Moreover the facies and fauna of the British Jurassic are hard to reconcile with the palaeolatitude of 66°N which derives from the present study if the magnetic field was axial, and faunal provinciality in Europe at the time is incompatible with the whole region having been inside the Jurassic arctic circle. We are driven to the conclusion that the geomagnetic field throughout the brief (ca. 1 million years) interval covered by our study was non-axial although, as one of our referees has remarked, if this is the case it is curious that the observed normal and reversed polarity groups are so close to 180° apart. Excursions of the field from its typical configuration for intervals of this duration have indeed been postulated recently from Jurassic palaeomagnetic studies elsewhere (Dr. E.A. Hailwood, personal communication).

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